

LSV PA1 report

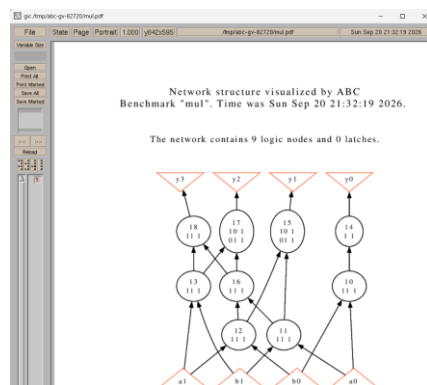
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2

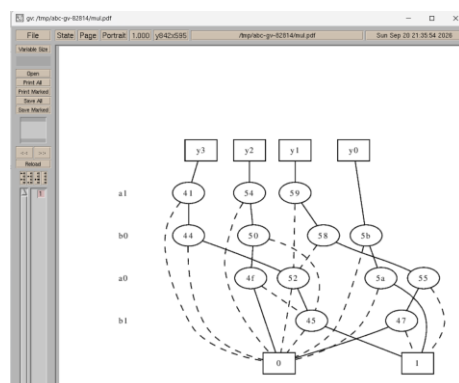
(b)

```
Baso@TheNasaMan: /mnt/c:/Users/Baso/OneDrive/Desktop/LSV/PA1/LSV-PA1$ ./abc
UC Berkeley, ABC 1.01 (compiled Sep 20 2026 21:21:43)
===== Command history =====
print_stats
strash
show mul.ps
sudo apt install gsfonts xfonts-base xfonts-75dpi xfonts-100dpi
source => abc.rc
read lsv/pa1/mul.blif
show
=====
abc 01> read lsv/pa1/mul.blif
abc 02> print_stats
mul
: i/o = 4/ 4 lat = 0 nd = 9 edge = 17 cube = 11 lev = 3
abc 02> show
abc 02> strash
abc 03> show
abc 03> collapse
abc 04> show bdd -g
```

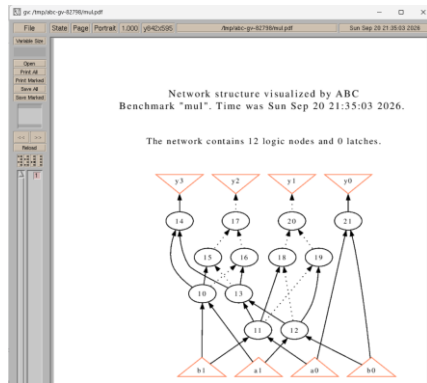
This is the screenshot for the terminal.



This is the screenshot for the first “show”.



This is the screenshot for the second “show”.



This is the screenshot for the “show_bdd -g”

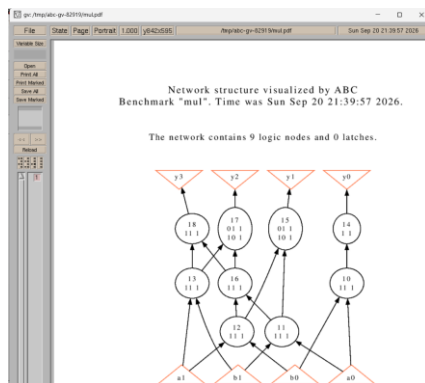
3

(a)

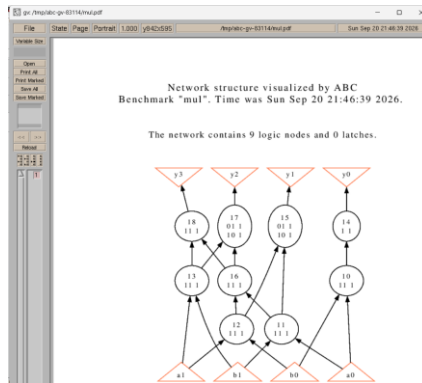
1.

```
NasaMan@TheNasaMan: /mnt/c:/Users/93er1/.vscode/LSV/pa1/LSV-PA$ ./abc
UC Berkeley, ABC 1.01 (compiled Sep 20 2026 21:21:43)
===== Command history =====
show mul.ps
sudo apt install gsfonts xfonts-base xfonts-75dpi xfonts-100dpi
print stats
strash
collapse
show_bdd -g
source -s abc.rc
read lsv/pa1/mul.blif
show
=====
abc 01> read lsv/pa1/mul.blif
abc 02> aig
abc 02> show
abc 02> strash
abc 03> show
```

This is the screenshot for the terminal.



This is the screenshot for the “aig”.



This is the screenshot for the “strash”.

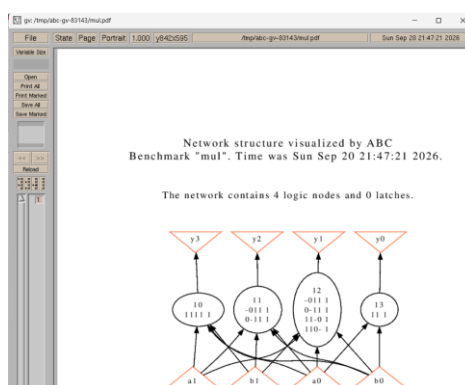
Findings:

aig keeps the original multi-level network topology and stores each node’s local function as a (possibly multi-gate) AIG fragment (nd=9, aig=12). strash flattens everything into one globally hashed AND-inverter graph (and=12). Structural hashing merges identical gates, so the representation is unique and level count can change.

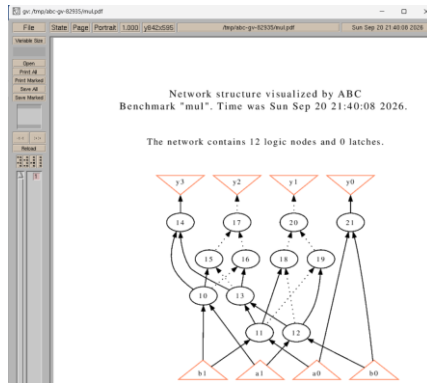
2.

```
NasaMan@TheNasaMan:~/mnt/c:/Users/93ert/.vscode/LSV/pa1/LSV-PA$ ./abc
UC Berkeley, ABC 1.01 (compiled Sep 20 2026 21:21:43)
===== Command history =====
show mul.ps
sudo apt install gsfonts xfonts-base xfonts-75dpi xfonts-100dpi
print stats
strash
collapse
show_bdd -g
source -s abc.rc
read lsv/pa1/mul.blif
show
=====
abc 01> read lsv/pa1/mul.blif
abc 02> bdd
abc 02> show
abc 02> collapse
abc 03> show
```

This is the screenshot for the terminal.



This is the screenshot for the “bdd”.



This is the screenshot for the “collapse”.

Findings:

bdd builds a local BDD for each internal node while preserving the multi-level structure (nd=9, lev=3). collapse builds global BDDs for the primary outputs only, eliminating internal nodes (nd=4 = four POs, lev=1).

(b)

```

Nava@nagTheNava@nag:/mnt/c/Users/936er1/.vscode/LSV/pa1/LSV-PA$ ./abc
UC Berkeley, ABC 1.01 (compiled Sep 20 2026 21:21:43)
===== Command history =====
show mul.ps
sudo apt install gsfonts xfonts-base xfonts-75dpi xfonts-100dpi
print_stats
strash
show_bdd -g
source -s abc.rc
read lsv/pa1/mul.blif
collapse
show
=====
abc 01> read lsv/pa1/mul.blif
abc 02> strash
abc 03> logic
abc 04> sop
abc 04> print_stats
mul : i/o = 4/ 4 lat = 0 nd = 12 edge = 24 cube = 12 lev = 4

```

This is the screenshot for the commands sequence in the terminal.